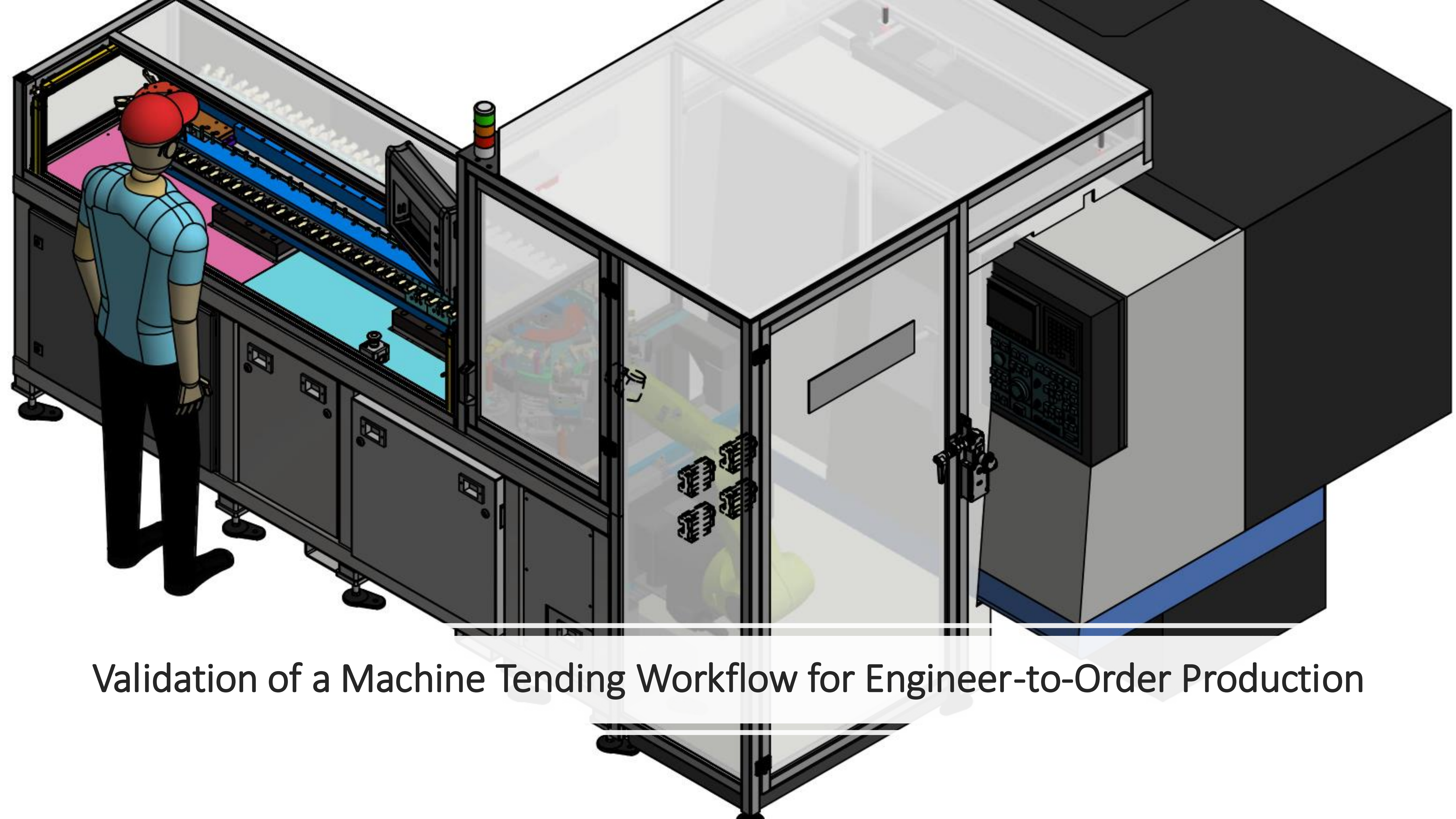


Validation of a Machine Tending Workflow for Engineer-to-Order Production

Privacy Notice for Participants

Purpose of the survey:
We are conducting this survey to collect anonymous feedback as part of our research on “validation of workflows in Engineer-to-Order environments”.

Use of MSForms:
The survey is designed to collect **anonymous responses by default**. You will not be asked to create an account or provide personal information to participate.

Data we collect:
Your responses to survey questions (multiple choice, open-ended, word clouds). No identifying information will be collected.

How your data will be used:
Your responses will be used solely for **research purposes**. All results will be presented in **aggregated form** so that no individual response can be identified.

Data retention:
Your responses will be stored securely within Mentimeter for **25 months** and will then be permanently deleted.

Legal basis:
Participation is voluntary, and by submitting your responses, you agree to the use of your data under **GDPR terms of use** for research purposes.

Your rights:
Under GDPR, you have the right to:

- Request access to your data (if applicable, although responses are anonymous).
- Withdraw your participation at any time before submission.
- Lodge a complaint with your national Data Protection Authority if you believe your data is being misused.
- If you have any questions about this privacy notice or the research project, please contact:

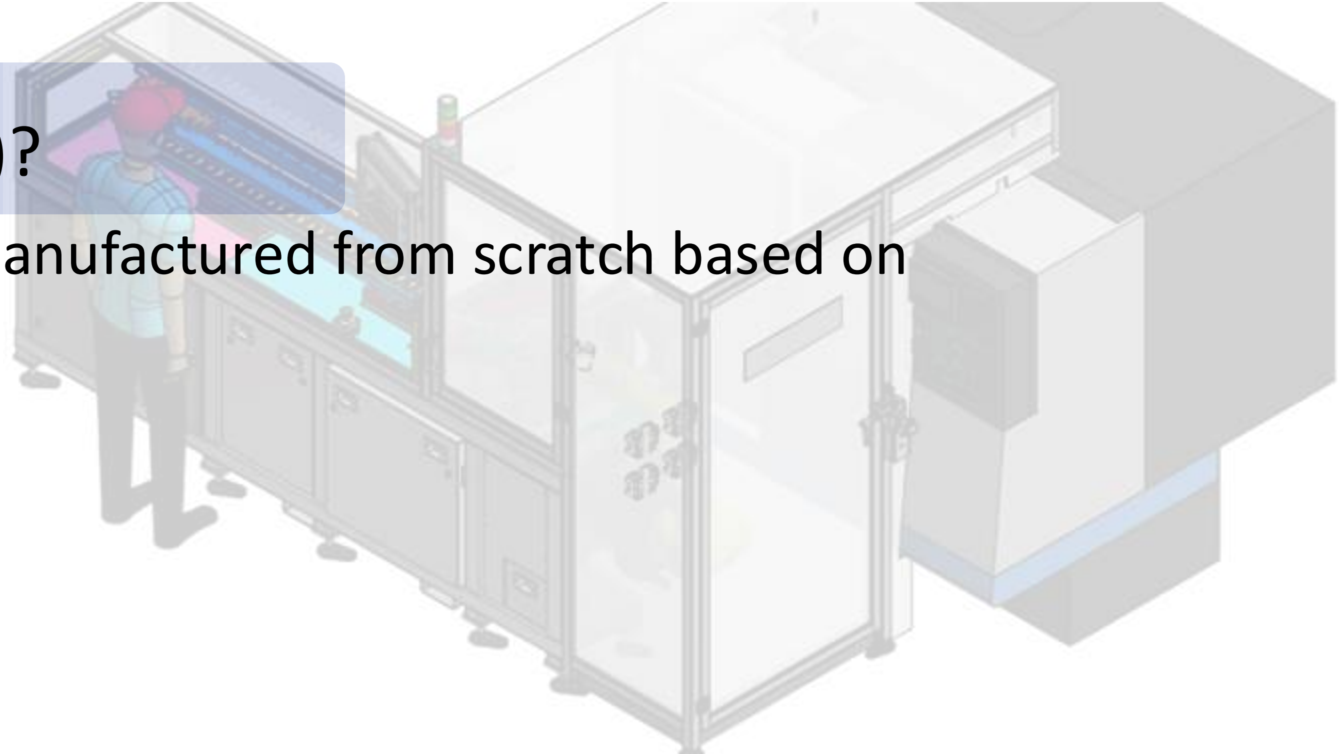
Beatriz Olarte, Politecnico di Milano, beatriz.olarte@polimi.it

Some Definitions

What do we understand for **Machine Tending**?

It is the process of automating the loading and unloading of a CNC machine. As an example, watch the following video.

https://www.youtube.com/watch?v=_uQlUCnNZvQ

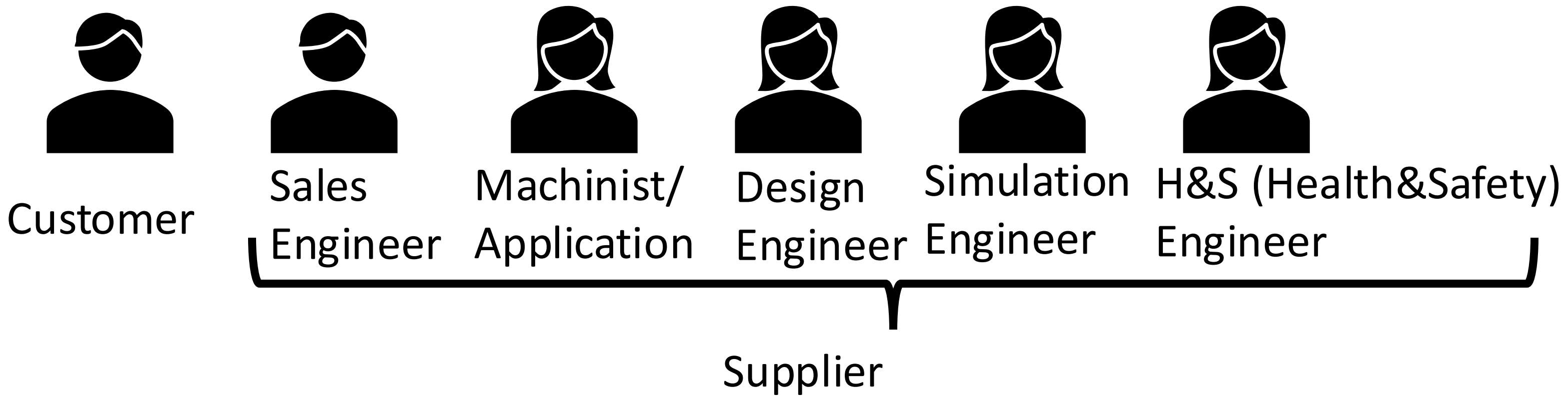


And for **Engineering-to-Order (ETO)**?

Products are designed, engineered and manufactured from scratch based on customer’s specific requirements.

How can you help?

We have identified 6 roles:



If you have worked in a company developing Engineering-to-Order products, related to robotic cells, machine tending, or machine tools and/or you have ordered them as a customer, you can provide your insights about the process, the length, the communication or any problems you have or had.

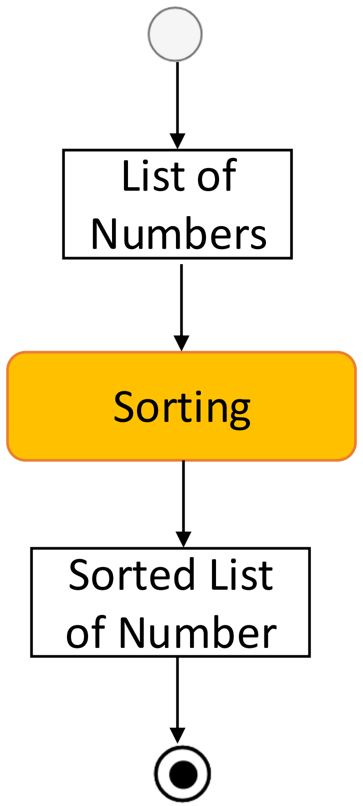
We are seeking for common sense and critical thinking that can highlight pitfalls and improvement on the current process

Workflow

We have represented the process of creating a quote for an ETO as a workflow.

In **computer science**, particularly in **algorithms and computation theory**, a *task* (or function, or problem) is abstracted as:

- **Input:** The data provided to the task.
- **Output:** The result produced by processing the input.



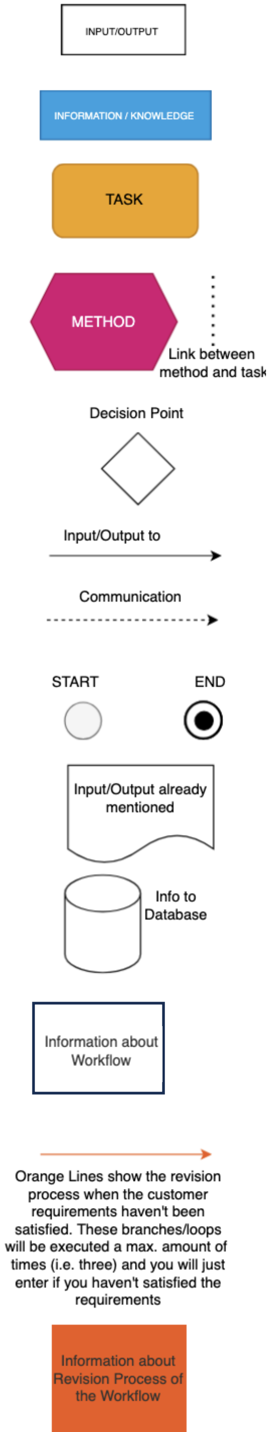
Example:
Sorting task:
- Input: A list of numbers.
- Output: A sorted list of those numbers.

This abstraction **ignores implementation details** and focuses on *what the task does* rather than *how* it does it.

There is a mathematical foundation for this but I think we can avoid it

Workflow

Legend:



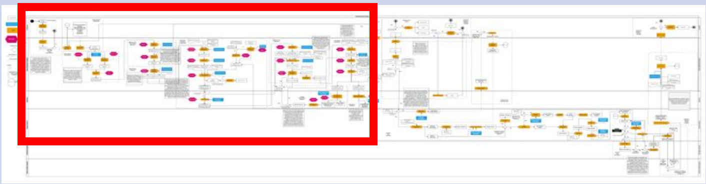
We have identified all the tasks that the different roles accomplish to obtain a quote for these types of products/projects.

Task

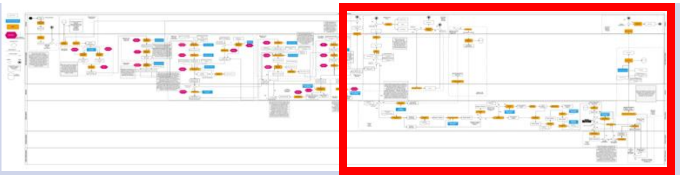
The tasks are identified in the yellow rectangle and always have at least an input and an output.

INFORMATION / KNOWLEDGE

Sometimes the input is in blue because we recognise the need for expert knowledge.



The initial part of the workflow involves selecting the best machine tool, cutting tools, and parameters for a given workpiece to produce. For this part, several methods to solve the task have been selected. However, we don't want to limit the **methods (pink hexagon),** as we aim to implement their digitalisation. Please **feel free to ignore them; this is not part of the validation**.



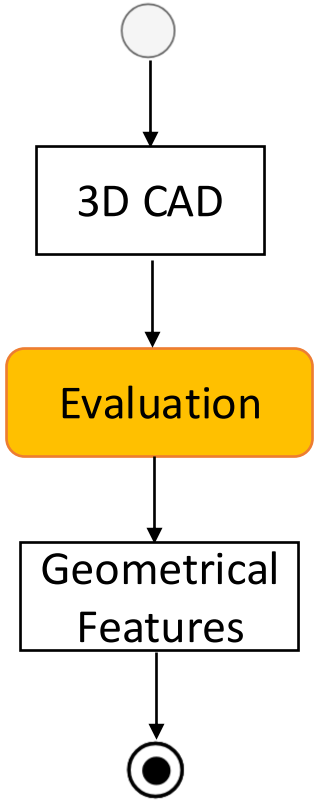
The second part is related to the design of the robotic Cell.

Comments

Orange Lines and Orange comments

Belong to the review process and enter into action when we haven't fully reached the throughput, but we close. Please don't confuse them with an infinite loop, as the idea is that you will apply a maximum number of times (i.e., two), and if no improvement is detected, we will go back to the customer.

Example:



The methods

METHODS

This is just if you are curious, for your information. You can ignore it!

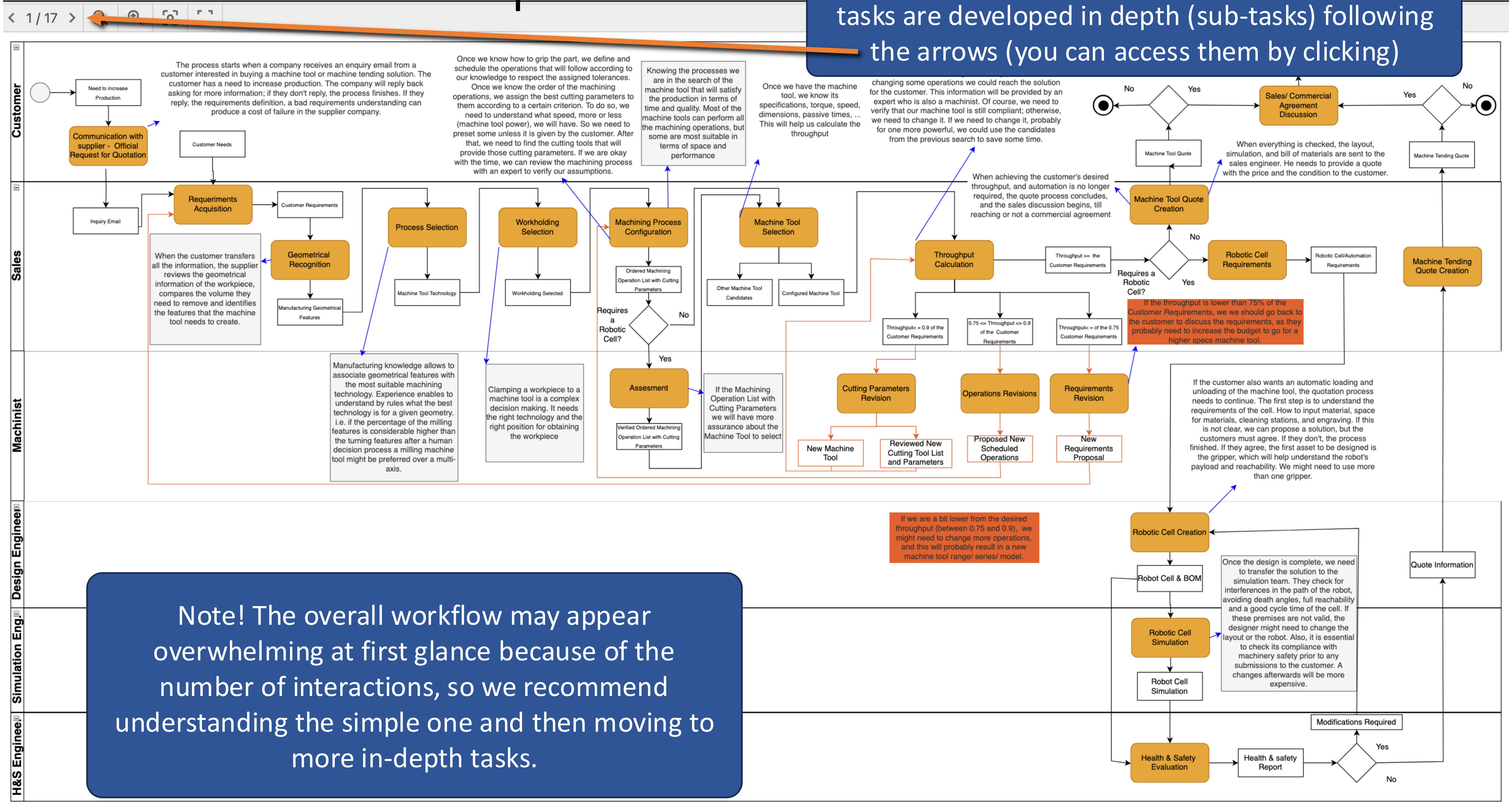
This is not very important, but for your acknowledgement, some of the methods that have been used to solve some of the tasks are taken from the CommonKADS methodology [1]. CommonKADS divides into:

- **Analytic Tasks:** Tasks dealing with **existing systems** (analysis, classification, diagnosis, monitoring).
- **Synthetic Tasks:** Tasks dealing with **creating new systems** (design, planning, scheduling).
- Some Tasks Examples:
  - **Classification Method -** Assign an object or situation to a predefined category. **Example**: An automated inspection system classifies metal parts as **"Pass" or "Fail"** based on surface defects measured by sensors.
  - **Assessment Method -** Evaluate objects or situations against criteria. Example: Assess process compliance with safety and GDPR requirements during machine tending.
  - **Association Method** - Link related data items without transformation. Example: Associating part numbers with their correct fixture IDs in a machine tending database.

[1] https://commonkads.org/

Workflow File Explanation

The Workflow is attached in the email, it consists of a high-level file on the first page. The individual tasks are developed in depth (sub-tasks) following the arrows (you can access them by clicking)



Note! The overall workflow may appear overwhelming at first glance because of the number of interactions, so we recommend understanding the simple one and then moving to more in-depth tasks.

Link to the survey

https://forms.office.com/e/RBC1whmZpW?origin=lprLink


Thank you for your time!